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DEPARTMENT OF INFORMATION TECHNOLOGY

Social Media Analytics

Unit-1

Topics

-World Wide Web

-Web 1.0

-Web 2.0

-Web 3.0

-Social Media

-Core Characteristics of Social Media

-Types of Social Media

-Social Networking Sites

-Using Facebook for Business Purposes

-Content Communities

World Wide Web

Definition: The World Wide Web (WWW), commonly called "the web," is a system of interlinked hypertext documents accessible via the Internet through web browsers.

- While the Internet provides the **infrastructure** (network of devices, cables, satellites, routers, and protocols), the **WWW provides content and services** in the form of websites, hyperlinks, and applications.

- Proposed by **Tim Berners-Lee** in **1989–1990** at CERN (European Organization for Nuclear Research).

- **Internet**: Hardware and protocols (TCP/IP) that connect computers.
- **WWW**: A service running on top of the Internet, using HTTP, HTML, and URLs to allow access to documents.
- Example: Without the Internet, the WWW cannot exist; but the Internet also supports services beyond the web (e.g., email, FTP, VoIP).

World Wide Web

<u>Components of Web:</u>

1)Hypertext
2)HTML
3)HTTP
4)URL

<u>Evolution of the Web</u>

1)Web 1.0-The Static Web
2)Web 2.0-The Social Web
3)Web 3.0-The Semantic and Intelligent Web

<u>How World Wide Web Works:</u>

1)User Request
2)DNS Resolution
3)HTTP/HTTPS Request
4)Server Processing
5)Transmission
6)Rendering

Example)

Type https://facebook.com → DNS returns edge IP (CDN/edge network).
Browser performs TLS handshake, sends GET / with cookies.
Servers/edge return bootstrap HTML and JS bundles; CDN supplies images/video.
Client React app runs, makes GraphQL/REST requests for feed data; WebSockets or push channels keep the feed real-time.
Analytics events (impressions, clicks) are sent by the client back to collection endpoints for aggregation.

World Wide Web

Types of Web Content:

1)Static Content

2)Dynamic Content

3)Multimedia Content

4)Hypermedia

Tools and Technologies of the World Wide Web:

1)Web Browser

2)Web Servers

3)Web Protocols

4)Markup Languages

5)Scripting Languages

6)Databases

Applications & Use Cases of the Web:

1)Information Access

2)E-Commerce

3)Communication

4)Social Media

5)Education

6)Entertainment

7)E-Governance

8)Banking and Finance

Benefits of the World Wide Web

1)Global Connectivity

2)Democratization of Information

3)Economic Growth

4)Social Empowerment & Knowledge Sharing

Web 1.0

Definition: **Web 1.0**, often referred to as the **“Read-Only Web”**, is the first generation of the World Wide Web, developed in the early 1990s and widely used until the early 2000s

It was **static in nature**, designed primarily for presenting information without user interaction or contribution.

Historical Background & Development of the World Wide Web:

1)Invention of the Web

2)Early Design Purpose

3)Timeline of Web 1.0

-1991-1993:Experimental Web

-1993-1996:Public Adoption Begins

-1996-2000:Commerical Explosion

-2000-2004:Gradual Transition

4)Primary Purpose of the Early Web

Web 1.0

5)Real World Examples of Early Web 1.0 Sites

<u>Key Characteristics of Web 1.0:</u>

1)Static Web Pages
2)Read-Only Content
3)One-Way Communication
4)Limited User Contribution
5)Page Design and Layout
6)Navigation
7)File-Based Hosting
8)Bandwidth Limitations

<u>Technical Foundation of Web 1.0:</u>

1)Core Technologies
-HTML, HTTP, URL

2)Browsers
-Mosaic(1993) & Netscape Navigator(1994)
-Internet Explorer(1995) & opera(1996)

3)Programming and Scripting

4)Web Servers

<u>Examples of Web 1.0 Websites:</u>

1)Static Corporate Websites
2)Early Portals
3)Educational and Governmental Sites
4)Online Encyclopedias
5)Personal Homepages

Web 1.0

Applications and Uses of Web 1.0:
1)Information Dissemination
2)Marketing and Promotion
3)Online Catalogs
4)News Publishing
5)Email Integration

Advantages of Web 1.0:
1)Global Accessibility
2)Low Complexity
3)Standardization
4)Foundation for Future Web

Limitations of Web 1.0:
1)Lack of Interactivity
2)Static Nature
3)Limited Multimedia
4)Poor User Engagement
5)No Personalization
6)Commercial Limitations

Societal and Business Impact of Web 1.0:
1)Shift from Print to Digital
2)Business Presence
3)Globalization of Knowledge
4)Foundation of E-Commerce
5)Cultural Impact

Case-Studies/Examples of Web 1.0:
1)Amazon(1995)
2)Yahoo Directory
3)CNN.com

Relevance: Still in Use, Educational Value, Historical Significance

Web 2.0

Definition:

Web 2.0 is the second generation of the World Wide Web that transformed the web from a static, one-way platform (Web 1.0) into a dynamic, interactive, and participatory ecosystem.

Unlike Web 1.0’s “read-only web,” Web 2.0 is the “read–write web,” where users are no longer passive consumers but active contributors.

Historical background and Evolution: Transition from Web 1.0 to Web 2.0:

1)Transition from Web 1.0 to Web 2.0
2)Societal Context Driving Web 2.0
3)Key Milestones in the Evolution

-1999: Blogger Launch
-2001: Wikipedia
-2003:MySpace and LinkedIn
-2004: Facebook and Flickr
-2005: You Tube
-2006-2010: Twitter, Instagram, Tumblr

Web 2.0

Core Characteristics of Web 2.0:

1)User Participation and User Generated Content
2)Two-Way and Many-to-Many Communication
3)Rich User Experience
4)Collaboration and Sharing
5)Social Networking
6)Tagging and Folksonomy
7)Openness and Participation

Tools and Technologies of Web 2.0:

1)Blogging Platforms
2)Podcasting
3)Social Bookmarking
4)Tagging Systems
5)Wikis
6)Social Networks
7)Video Sharing
8)AJAX and Rich Internet Applications(RIA)

Applications and Use Cases of Web 2.0:

1)Social Media Platforms
2)Collaborative Knowledge
3)E-Commerce and Reviews
4)Education
5)Business and Marketing
6)Entertainment
7)Citizen Journalism

Web 2.0

Advantages of Web 2.0:
1)Active User Participation
2)Community Building
3)Ease of Publishing
4)Collaboration
5)Economic Opportunities
6)Accessibility

Limitations and Challenges of Web 2.0:
1)Information Overload
2)Misinformation and Fake News
3)Privacy Concerns
4)Cybersecurity Risks
5)Digital Divide
6)Monetization Issues

Social and Cultural Impact of Web 2.0:
1)Democratization of Content
2)Rise of Social Movements
3)Transformation of Media
4)Global Communities

Business and Economic Impact of Web 2.0:
1)Shift to Digital Marketing
2)New Business Models
3)Customer Engagement
4)E-Commerce Evolution

Case-Studies: Wikipedia, You Tube, Facebook, Amazon Reviews

Relevance- Still Dominant, Foundation for Web 3.0, Influence on Culture and Business

Web 3.0

Definition: Web 3.0 is the third generation of the World Wide Web, envisioned as a semantic, intelligent, and connected web where data, applications, and people are linked more meaningfully.

It builds upon Web 2.0's interactivity and user participation, but introduces a smarter, machine-readable layer that enables software and systems to interpret, process, and connect data autonomously.

Historical Background and Evolution of the Web:

1)From Web 1.0 to Web 3.0

2)Introduction of the Semantic Web

3)Technological Drivers

4)Timeline and Key Milestones

Web 3.0

Key Characteristics of Web 3.0:

1)Semantic Understanding

2)Ubiquity and Connectivity

3)Decentralization

4)Artificial Intelligence Integration

5)Personalization

6)Interoperability

7)3D and Immersive Web

Technologies Underpinning Web 3.0:

1)Semantic Web

2)Linked Data

3)Artificial Intelligence and Machine Learning

4)Natural Language Processing

5)Blockchain Technology

6)3D Graphics, AR/VR

Web 3.0

The Semantic Web-Central to Web 3.0

The Semantic Web is an extension of the World Wide Web that structures data in a way that is machine-readable, allowing computers to understand, interpret, and process information automatically.

Traditional web (Web 2.0) is considered a “web of documents”, where content is mainly for human consumption.

The Semantic Web transforms this into a “web of data”, where information is linked, meaningful, and interpretable by machines, allowing for context-aware responses and intelligent automation.

Key Technologies: RDF,OWL,SPARQL

Example)Searching for a Keyword such as Paris might show result relevant to User’s intent in Web 3.0.

Web 3.0

Applications and Use Cases of Web 3.0:

1)Search Engines
2)Virtual Assistants
3)E-Commerce
4)HealthCare
5)Finance and Banking
6)Social Media
7)Education
8)Entertainment
9)Metaverse

Advantages of Web 3.0:

1)Smarter and More Efficient Web
2)Personalized Experience
3)Interoperability
4)Decentralization
5)Trust and Security
6)Innovation and New Business Models

Limitations and Challenges of Web 3.0:

1)Complexity of Implementation
2)High Resource Requirements
3)Data Privacy Concerns
4)Digital Divide
5)Scalability Issues
6)Unclear Governance

Social and Cultural Impact of Web 3.0:

1)Shift in Data Ownership
2)Empowerment of Users
3)Cultural Integration
4)New forms of Community

Web 3.0

Business and Economic Impact of Web 3.0:

1)E-Commerce
2)Marketing
3)Decentralized Finance(DeFi)
4)Token Economy
5)Disruption of Traditional Models

Case Studies/Examples of Web 3.0:

1)Google Knowledge Graph
2)IBM Watson
3)Ethereum(Blockchain)
4)Brave Browser
5)Decentraland

Relevance: Emerging Reality, Industry Focus, Transition in Progress

Social Media

Definition:

Social media refers to Internet-based platforms and tools that enable users to create, exchange, and share content (text, audio, video, images) in an interactive and many-to-many communication environment.

It is an application of Web 2.0 philosophy, where users are empowered to contribute, collaborate, and co-create content, unlike the earlier Web 1.0 static model.

Nature and Scope of Social Media:

1)Beyond Famous Platforms

-Not Limited to Popular Sites

-Proprietary and Purpose Built Platforms

-Global and Local Reach

-Dynamic and Participatory Ecosystem

Social Media

2)Forms of Content on Social Media

-Text Based Content
-Audio Content
-Video Content
-Images and Graphics
-Hybrid and Multimedia Content

Nature of Communication on Social Media:

1)Many to Many Communication Model
2)Role of Users as Creators, Distributors and Consumers
3)Real Time and Contextual Communication
4)Collaborative and Participatory Nature

Scope and Influence:

- Social media is not only a platform for personal communication but also a tool for businesses, education, governance, activism, entertainment, and cultural exchange.
- It influences consumer behavior, marketing strategies, political movements, and social trends, making it a key component of contemporary society.

Social Media

The Evolution of Social media can be divided into distinct phases reflecting technological advancements, user behavior changes and broader societal trends.

1)Early Internet Communities(Pre Web 2.0)
-Platforms
-Nature of Interaction
-Interactivity
-User Experience
-Significance

2)Web 2.0 Transformation(2000s)
-Launch of Key Platforms
-Shift in Paradigm
-User Participation
-Collaborative Features
-Impact

3)Mobile and Multimedia Expansion(2010s)
-Mobile First Platform
-Multimedia Dominance
-Real-Time Engagement
-Algorithmic Personalization
-Economic Impact

4)Present and Future(2020s Onwards)
-Integration of AI and Machine Learning
-Short Form and Ephemeral Content
-Live Streaming and Interactivity
-Decentralized Social Networks
-Societal and Cultural Shfits

Social Media

Types of Social Media Platforms:

1)Social Networking Sites
2)Media Sharing Networks
3)Discussion Forums
4)Collaborative Projects
5)Blogging and Microblogging Platforms
6)Messaging and Community Apps
7)E-Commerce Integration

Applications and Uses of Social Media:

1)Personal Communication and Networking
2)Business and Marketing
3)Education and Knowledge Sharing
4)Entertainment and Leisure
5)Politics and Activism
6)Professional Development

Advantages of Social Media:

1)Accessibility and Low Barriers
2)Global Reach
3)Community building
4)User Empowerment
5)Real Time Information
6)Cost-Effective Marketing
7)Enhanced Collaboration and Learning
8)Cultural and Social Impact

Challenges and Issues in Social Media:

1)Misinformation and Fake News
2)Privacy Concerns
3)Addiction and Mental Health Issues

Social Media

4)Polarization and Echo Chambers
5)Digital Divide
6)Reputational and Legal Risks
7)Cyber bullying and Harassment

Social , Cultural and Economic Impact of Social Media:

1)Social Impact:
-Transformation of Communication Habits
-Strengthening of Virtual Communities
-Cross-Cultural Exchange

2)Cultural Impact:
-Creation of New Digital Cultures
-Blurring of Personal and Professional Identities
-Influence on Public opinion and Civic Engagement

3)Economic Impact:
-Rise of the Creator Economy
-Transformation of Marketing Strategies
-E-Commerce Integration
-Gig Work and Freelancing Opportunities
-Business Intelligence and Analytics

Relevance of Social Media:

-Dominant Online Activity
-Essential for Businesses
-Key in Politics and Governance
-Integration with Emerging Technologies

Core Characteristics of Social Media

1)<u>Social Media is Many-to-Many</u>
-Shift from One-Way to Many-to-Many Communication
-Interactivity at Scale
-Decentralized Participation
-Global Conversations and Connectivity
-Implications:

2) <u>Social Media is Participatory</u>
-Active User Involvement
-Feedback and Dialogue Mechanisms
-User Agency and Control:
-Collaborative Knowledge Contribution
-Implications:

3)<u>Social Media is User Owned</u>
-Content Creation by Users
-Value is user Driven
-Intellectual Property Ownership
-User driven Trends and Virality
-Implications

4)<u>Social Media is Conversational</u>
-Ease of Real Time Conversations
-Many to Many Conversations at Scale
-Two way interactions with Institutions
-Collaborative and Community Conversations
-Implications

Core Characteristics of Social Media

5)Social Media Enables Openness
-Access to Information
-Low Barriers to Entry
-Transparency and Visibility
-Information Democratization
-Implication

6)Social Media enables Mass Collaboration
-Collective Participation
-Crowd Sourcing and Crowd Funding
-Hashtag Campaigns and Advocacy
-Crisis Management and Disaster Relief
-Implications

7)Social Media is Relationship Oriented
-Focus on Building Connections
-Sustaining Long Distance Ties
-Networking Opportunities
-Community building
-Implications

8)Social Media is Free and Easy to Use
-Low Cost Accessibility
-User Friendly Interfaces
-Scalable Use across contexts
-Business Model of Platforms
-implications

Core Characteristics of Social Media

9)<u>Broader Implications of Core Characteristics:</u>
-Transformation of Media Ecosystem
-Power Shifts To Users
-Impact on Business and Society
-Technology-Culture Fusion

Types of Social Media

- Social media platforms can be classified in several ways—by their purpose, by their content type, or by their user base.
- Broadly, social media can be categorized into two types based on authentication and access mechanisms: Internet-Based Social Media and Smartphone-Based Social Media.

1)**<u>Internet based Social Media</u>**

<u>Key Characteristics:</u>

-Authentication via Email ID
-Cross Device Accessibility
-Global Connectivity
-Support for Communities and Groups
-Longevity and Stability of Accounts
-Cultural and Social Influence
-Content Versatility
-Rich User Profiles
-Corporate and Business Relevance
-Analytics and Insights
-Integration with Other Services
-Examples and Implications

Types of Social Media

2)**Smartphone based Social Media**

Key Characteristics:

-Authentication via Mobile Numbers

-Mobile First Nature

-Integration with Mobile Features

-Location based Services

-Instant Accessibility

-Personalization through Mobile Data

-Ease of Registration and Use

-Casual and Contextual Usage

-Push Notifications for Engagement

-Lightweight and Fast Interaction

-Privacy and Security Aspects

Social networking Sites

Definition: Social Networking Sites (SNS) are online platforms designed to enable users to create profiles, establish connections, and interact with others for personal, professional, or communal purposes. They serve as digital ecosystems where individuals and organizations maintain relationships, share content, and build communities while leveraging technology to expand social interactions beyond physical boundaries.

Features:

-Relationship Oriented Nature
-Profile Creation as Digital Identity
-Connections and Links
-Network Navigation
-Multimedia Support
-Community Formation
-Algorithmic Personalization
-Role in Information Sharing
-Global Reach and Accessibility
-Examples

Social networking Sites

Types of Social Networking Sites:

- General Social Networks
- Professional Networks
- Interest Based Networks
- Microblogging Platforms
- Discussion Forums and Communities
- Business and Collaboration Networks

Advantages:

- Global Connectivity
- Business Growth
- Educational Value
- Creativity and Expression
- Information Sharing
- Community Building
- Career Opportunities

Disadvantages/Challenges:

- Privacy Issues
- Cyberbullying
- Addiction and Time Wastage
- Misinformation and Fake News
- Online Scams and Frauds
- Mental Health Concerns

Using Facebook for Business Purposes

Businesses use Facebook effectively in the below mentioned ways:

1)Marketing through Pages

2)Community and Networking Hubs

3)Targeted Brand Awareness

4)Content Variety

5)Customer Feedback Mechanism

6)Engagement Strategy Design

7)Reputation and Security Risks

8)Analytics and Insights

9)Cost-Effective Marketing

10)Examples of Usage

Content Communities

- Definition: **Content communities** are online platforms that center around the creation, sharing, and consumption of specific types of digital content such as videos, images, audio, or presentations.
- Unlike social networking sites that primarily focus on relationships, content communities emphasize **user-generated media** as the main driver of interaction, engagement, and value.

Features:

1)Content-Centric Design

2)Search and Discovery Features

3)User Generated Participation

4)Community Interactions

5)Global Visibility

6)Content Longevity

7)Professional and Creative Showcase

8)Integration with Other Platforms

Content Communities

9)Importance in the Digital Economy

10Examples

Types of Content Communities:

-Video Sharing

-Photo Sharing

-Music and Audio Sharing

-Document and Presentation Sharing

-Art and Design Communities

-Knowledge Sharing Communities

Advantages:

-Free and Open Access

-User Empowerment

-Global Exposure

-Interactive Engagement

-Learning and Career Opportunities

Disadvantages:

-Plagiarism Issues

-Misinformation

-Cyberbullying and Negative Comments

-Privacy Risks and Monetization Inequalities

Blogs

Definition: A blog is an online platform, often maintained by individuals or organizations, that functions like a digital journal or website where content is regularly updated in the form of posts.

Features:

1)Personal and Organizational Spaces

2)Flexibility of Content

3)Interactivity with Readers

4)Chronological Archives

5)Subscription and Notifications

6)Focused Themes or Niches

7)Accessibility and Cost Effectiveness

8)Authorial Voice and Identity

9)Global Reach

10)Integration with Other Media

Blogs

Types of Blogs:

1)Personal Blogs
2)Professional Blogs
3)Niche Blogs
4)News Blogs
5)Educational/Academic Blogs
6)Guest Blogs
7)MicroBlogs

Main Purpose of Blogs:

1)Information Sharing
2)Self-Expression
3)Education and Learning
4)Marketing and Branding
5)Community Building
6)Revenue Generation

Advantages:

-Ease of Use
-Improves Communication
-Enhances Online Presence
-Knowledge Sharing
-Monetization

Disadvantages: Time Consuming, Competition, Misinformation, Plagiarism, Maintenance Costs.

Micro-Blogging

Definition: Microblogging is a form of blogging that emphasizes **brevity and frequency**, allowing users to share short updates, typically limited in characters, along with links, images, or videos.

Key Characteristics of MicroBlogging:

1)Short and Quick Posts
2)Real Time Updates
3)Public Visibility
4)Follower based System
5)Hashtags and Mentions
6)Multimedia Integration
7)Instant Interaction

Purpose:

1)Quick Information Sharing
2)Social Interaction
3)Business and Marketing
4)Education and Awareness
5)Public Communication
6)Event Updates and Live Coverage

Advantages: Speed and Simplicity, Wide Reach, Community Building, Marketing and Branding, Real Time Communication, Accessibility
Disadvantages: Limited Message Length, Misinformation Spread, Privacy Risks, Cyberbullying.

Online Collaborative Projects

1)WIKIS and Collaboration

- Wikis function as collaborative environments where multiple users can contribute, refine, or even delete information, ensuring that knowledge is not static but continuously evolving.
- Platforms like Wikipedia, Google Sites, and Wikispaces demonstrate how global participation transforms a simple repository into a dynamic knowledge hub.
- Version control is integral to wikis, meaning every edit is tracked and stored, so past versions can always be retrieved if needed.
- Edit histories create accountability, since each contribution is linked to a specific user, discouraging misuse while encouraging responsibility.
- Open participation allows inclusivity, enabling experts, amateurs, and enthusiasts alike to shape the body of knowledge.
- Collective intelligence thrives in this model, where the combined insights of many individuals often surpass the expertise of a single source.
- Wikis encourage continuous updates, which makes them particularly effective in documenting fast-changing domains such as technology, medicine, or education.
- The collaborative design fosters teamwork, as contributors learn to negotiate, discuss, and validate information before it becomes widely accepted.

Folksonomies/Tagging

Definition: **Folksonomy** is a collaborative, user-driven system for classifying and organizing online content. Unlike a formal, expert-designed taxonomy, a folksonomy emerges from the "folks" (the users) freely applying descriptive tags or hashtags to content.

This practice, also known as social tagging, is a rich source of unstructured data for social media analysis.

Characteristics of Folksonomies:

1)User-Generated

2)Non-Hierarchical

3)Collaborative Nature

4)Dynamic and Evolving

5)Keyword Based Retrieval

Types of Folksonomies:

1)Broad Folksonomy

2)Narrow Folksonomy

Importance of Folksonomies/Tagging:

1)Improved Searchability

2)Organization of Information

3)User Participation

4)Trend Identification

5)Personalization

Virtual Worlds

Definition: Virtual worlds are **computer-based, simulated environments** where users can interact with each other and the digital surroundings through **avatars** (digital representations of themselves). These worlds are **persistent**, meaning they continue to exist and evolve even when users are offline.

Characteristics of Virtual Worlds:

1)3D Environment
2)Avatars
3)Persistence
4)Interactivity
5)Community and Socialization
6)User-generated Content

Types of Virtual Worlds:

1)Social Virtual Worlds
2)Gaming Virtual Worlds
3)Educational Virtual Worlds

Virtual Worlds

<u>Uses and Applications:</u>

1)Social Interaction
2)Education and Training
3)Business and Marketing
4)Entertainment
5)Virtual Economy

<u>Advantages:</u>

- Enables global interaction without physical boundaries.
- Encourages creativity and innovation through user-generated content.
- Provides immersive learning and training experiences.
- Builds strong online communities and collaborations.

<u>Limitations and Challenges:</u>

- Technical Requirements
- Privacy and Security Issues
- Addiction and Overuse
- Regulation and Ownership

Purpose Built Platforms

Definition: Purpose-built platforms are specialized social media or online platforms that are designed for a specific purpose, audience, or type of activity rather than for general social networking.

Unlike broad platforms such as Facebook or Twitter, which serve multiple functions (communication, sharing, entertainment, etc.), purpose-built platforms are created with a particular focus, such as professional networking, photo sharing, music streaming, fitness tracking, or education.

Characteristics of a Purpose Built Platform:

1)Specific Function or Goal
2)Targeted Audience
3)Focused Features and Tools
4)Community Building
5)High Engagement Quality

Purpose Built Platforms

Uses and Applications:

1)Professional Growth
2)Learning and Skill Development
3)Creative Expression
4)Entertainment and Media Consumption
5)Health and Lifestyle Management

Advantages:

1)Focused User Experience
2)Relevant Community
3)High Engagement and Productivity
4)Professional and Educational Value

Limitations:

1)Limited Scope
2)Smaller Audience base
3)Dependence on Niche Interest
4)Data Privacy Concerns

Mobile Apps

Definition: Mobile Apps are Software Programs designed to run on mobile devices such as Smartphones and Tablets. They are created to perform specific tasks such as communication, shopping, entertainment, banking or learning and are available for download through app stores like Google Play or Apple App Store.

Characteristics of Mobile Apps:

1)Designed for Mobile Devices

2)User Friendly Interface

3)Internet Connectivity

4)Personalization

5)Integration with Device Features

6)Notifications and Updates.

Advantages: Convenience, Speed and Performance, offline Access, Personalization, Enhanced Functionality.

Disadvantages: Storage and Memory Use, Frequent Updates, Compatibility Issues, Privacy and Security Risks, Internet Dependency.